

Aurick Anwar

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EDUCATION

McMaster University

Bachelor of Engineering: Engineering Physics

Hamilton, Ontario

Expected 2029

WORK EXPERIENCE

Software Engineer

McMaster Drone Team

Hamilton, Ontario

June 2026 – Present

- Developed an autonomous drone pipeline using **YOLOv11**, **OpenCV**, and **PyTorch**, achieving **30+ FPS** real-time object detection with **90%+ mAP** in **ArduPilot** simulated environments.
- Simulated and validated **100+** autonomous flight missions in **NVIDIA Isaac Sim**, testing navigation and obstacle avoidance in different environments before deploying.
- Built **10+ ROS2** nodes, enabling a **<100 ms** communication between perception, planning, and control systems.

Software Engineering Intern

HermesAI

Toronto, Ontario

May 2026 - Present

- Built **React** interfaces for document upload and AI-powered case extraction, supporting legal files exceeding **1,000 pages**.
- Built an **LLM** chatbot feature with **LangChain** that parses, chunks, and extracts case law from legal documents, cutting manual processing time by **50%** for Canadian law firms.
- Reduced document processing latency by **30%** by offloading file processing to asynchronous **Node.js**.

PERSONAL PROJECTS

🔗 Autonomous Self-Driving Vehicle with CARLA [CARLA, YOLOv11, PyTorch, OpenCV]

- Developed an end-to-end autonomous driving pipeline in the **CARLA** simulator integrating **YOLO-based object detection** and vehicle control, achieving a real-time simulation at **~20-30 FPS**.
- Wrote to a dataset with **5+ features** (steer, brake, speed, acceleration, throttle), generating **25k+** sample datasets to train a **PyTorch** model.
- Used **CrossEntropyLoss**, achieving **~88.5%** prediction directional accuracy for precise automation navigation.

🔗 Breast Cancer Cell Detection [PyTorch, Roboflow, YOLOv11, Grad-Cam, CNNs]

- Fine-tuned **Roboflow YOLOv11** tech stack on **5,400+** images to localize breast cancer regions with bounding box detection across **benign** and **malignant** classes.
- Built a custom **3-layer PyTorch CNN** classifier achieving **95.31%** confidence on malignant tumor classifications using **CrossEntropyLoss** and **Adam optimization**.
- Implemented **Grad-CAM**, generating **spatial heatmaps** that visually highlight the tumor regions driving each prediction.

🔗 VentBuddy [LangChain, Ollama, Speech-Recognition, Flask]

- Built a speech-recognition feature enabling users to voice their problems, transcribing input in real time with **95% accuracy**.
- Deployed a conversational engine with **LangChain** and **Ollama**, maintaining full session context locally with **<3s** response times.
- Generated an end-of-session summary document report, summarizing key themes and emotional patterns, reducing manual reflection time by **70%**.

Technical Skills

Programming Languages: Python, C++, Javascript, CSS, HTML, MATLAB,

ML and Computer Vision: PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLOv11, LangChain, Mediapipe

Robotics and Simulation: ROS, RVIZ, Gazebo, SLAM, Nvidia Isaac Sim, ArduPilot

Electronics and Microcontrollers: Arduino, Raspberry Pi, ESP32

3D Design and CAD: Solidworks, Inventor, Fusion360, AutoCAD

Web Development: Node.js, React

Tools and OS: Linux, Git, Docker